

Polymerase chain reaction (PCR)

The power of exponential growth

Learning goals

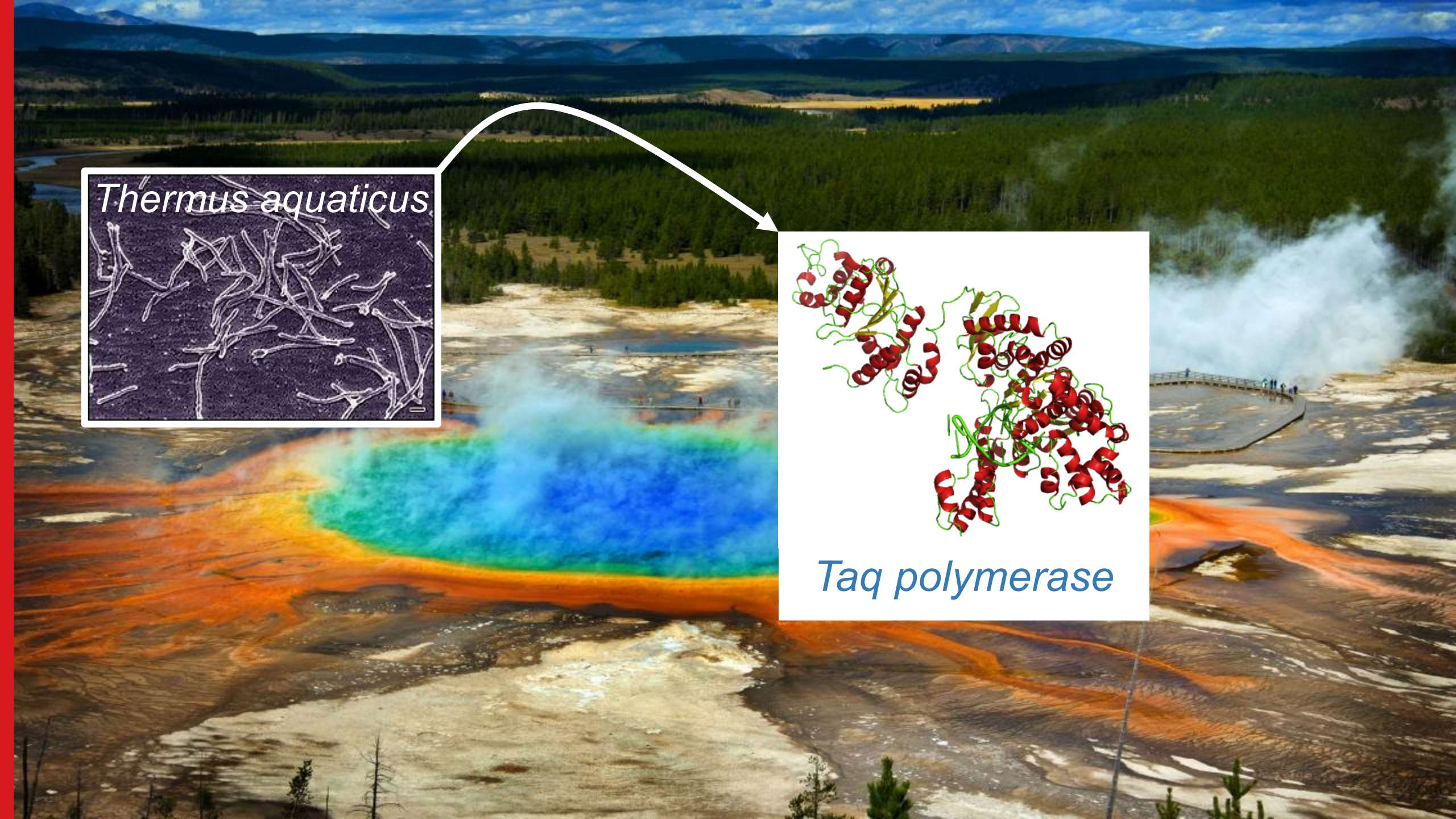
- Understand the principles and main steps of PCR
- Explore applications of PCR
- Interpret real-time PCR results



Let’s design a method to amplify DNA

Is there a tool to copy a DNA fragment?

What steps should I follow to copy a fragment of DNA?


Thermus aquaticus
Taq polymerase

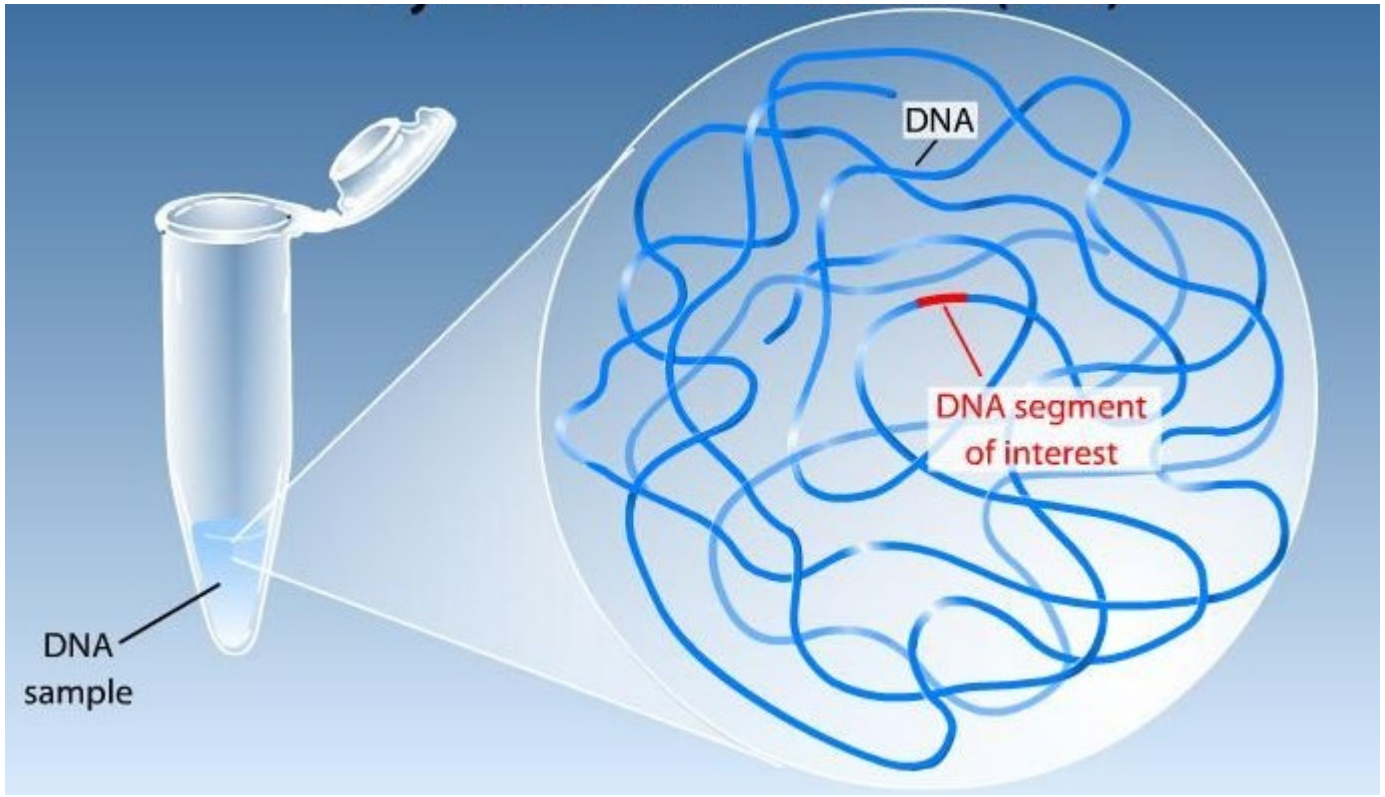
A brief history of PCR



- **1983** Dr. Kary Mullis developed PCR while working at Cetus Corporation
- **1985** The first publication of PCR appears in Science
- **1986** Purified *Taq polymerase* is first used in PCR
- **1988** PCR using a thermostable Taq DNA polymerase is published
- **1988** Published studies use PCR to amplify cDNA generated by reverse transcription
- **1991** Roche buys rights for Taq and PCR patents from Cetus for $300 million
- **1992** A fluorescent DNA-binding dye is applied laying the foundation for qPCR
- **1993** Kary Mullis shares Nobel Prize in Chemistry for conceiving PCR technology

Objective

To generate millions of copies of a specific fragment of DNA by using an enzymatic method based on DNA replication, and the repetition of these three steps:



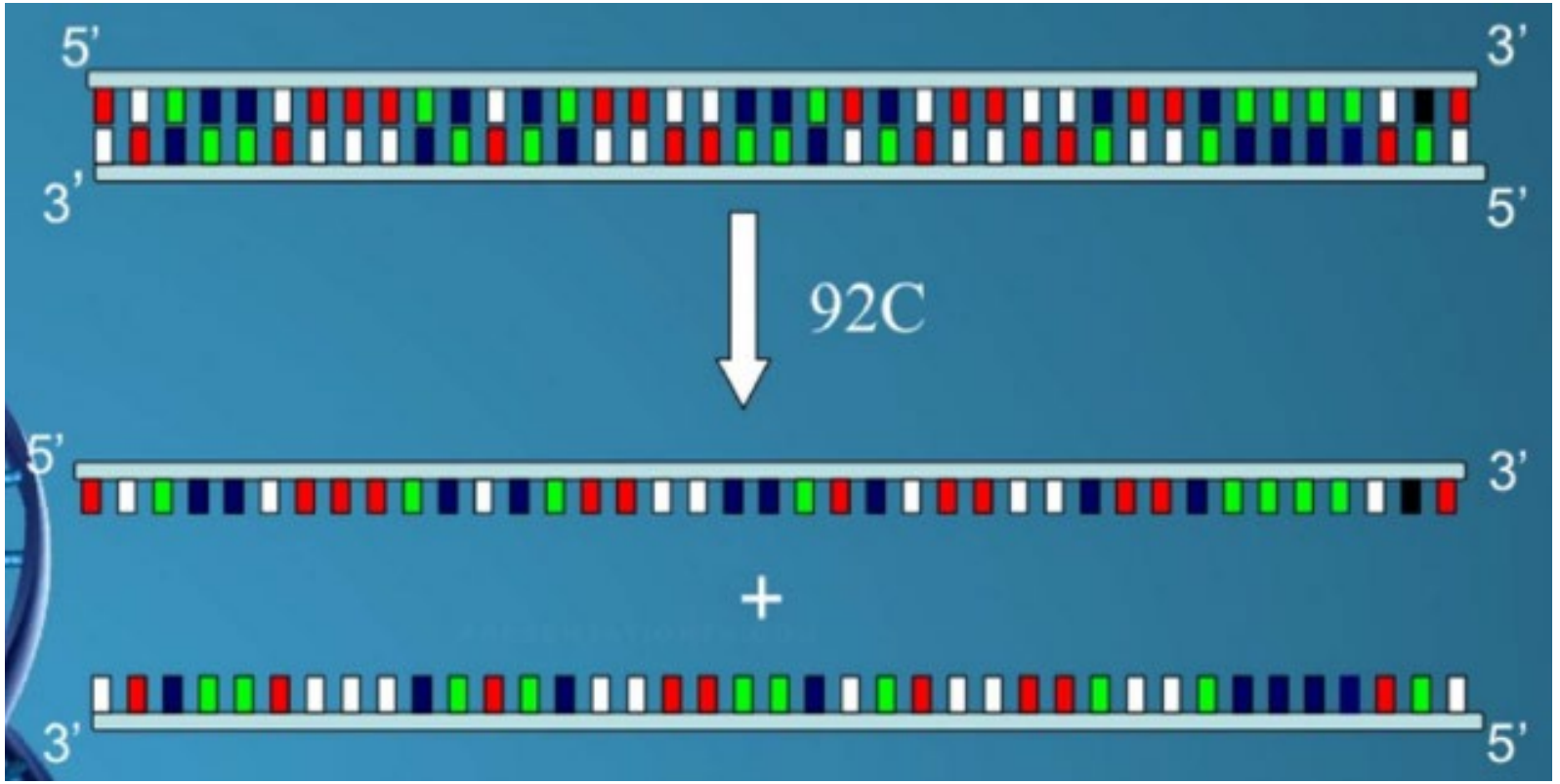
1. Denaturation
2. Primer annealing
3. Extension

Necessary knowledge

Basic reagents needed

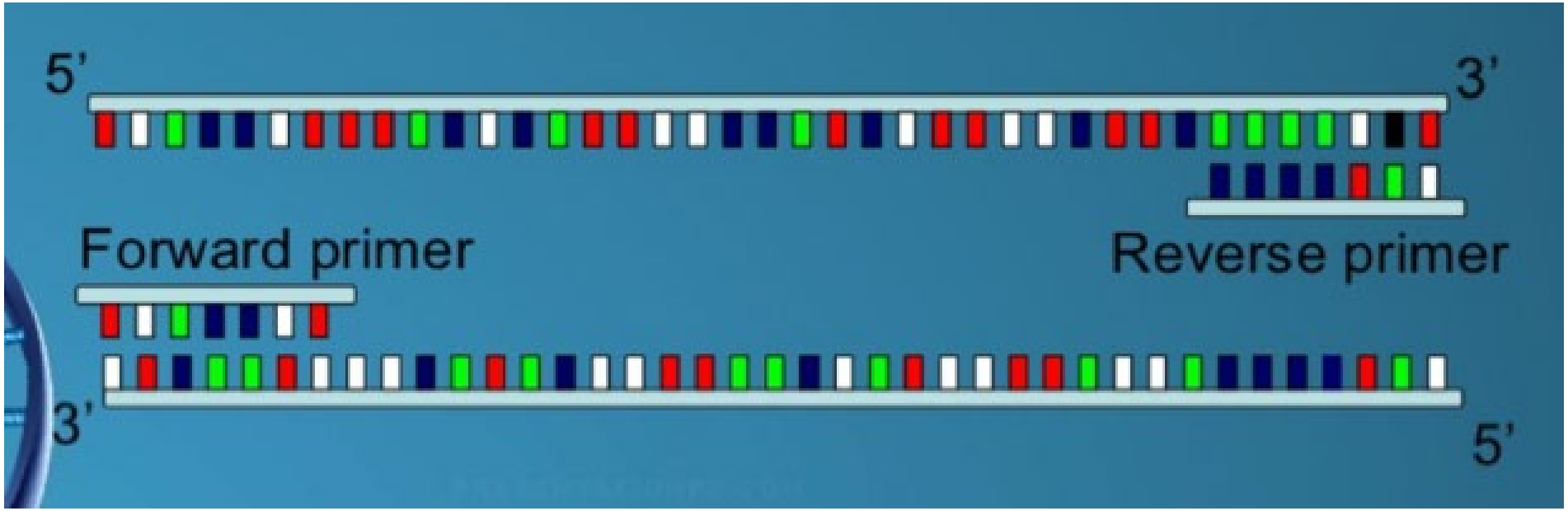
Equipment

Step 1. Denaturation



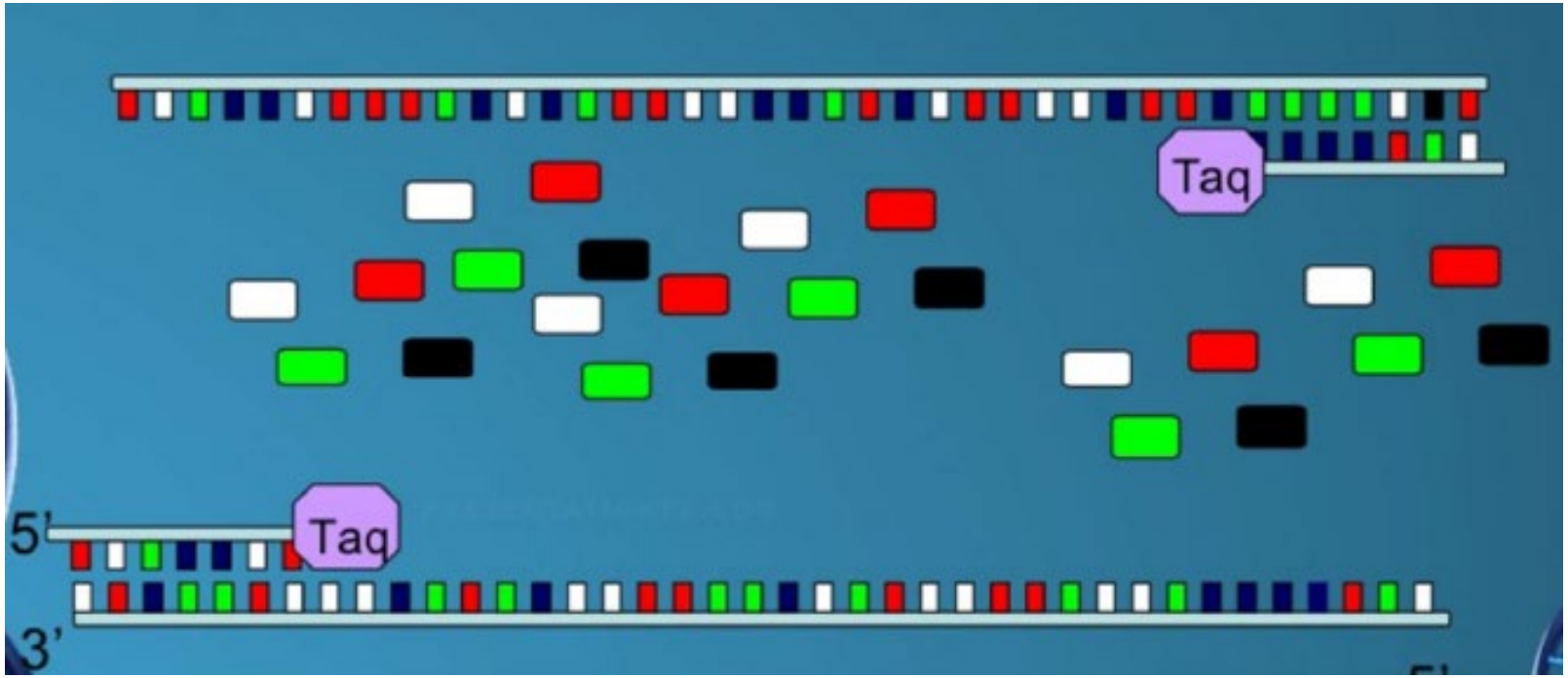
- Sample is heated to 92-95 ºC
- Double-stranded DNA melts into single-stranded DNA

Step 2. Annealing



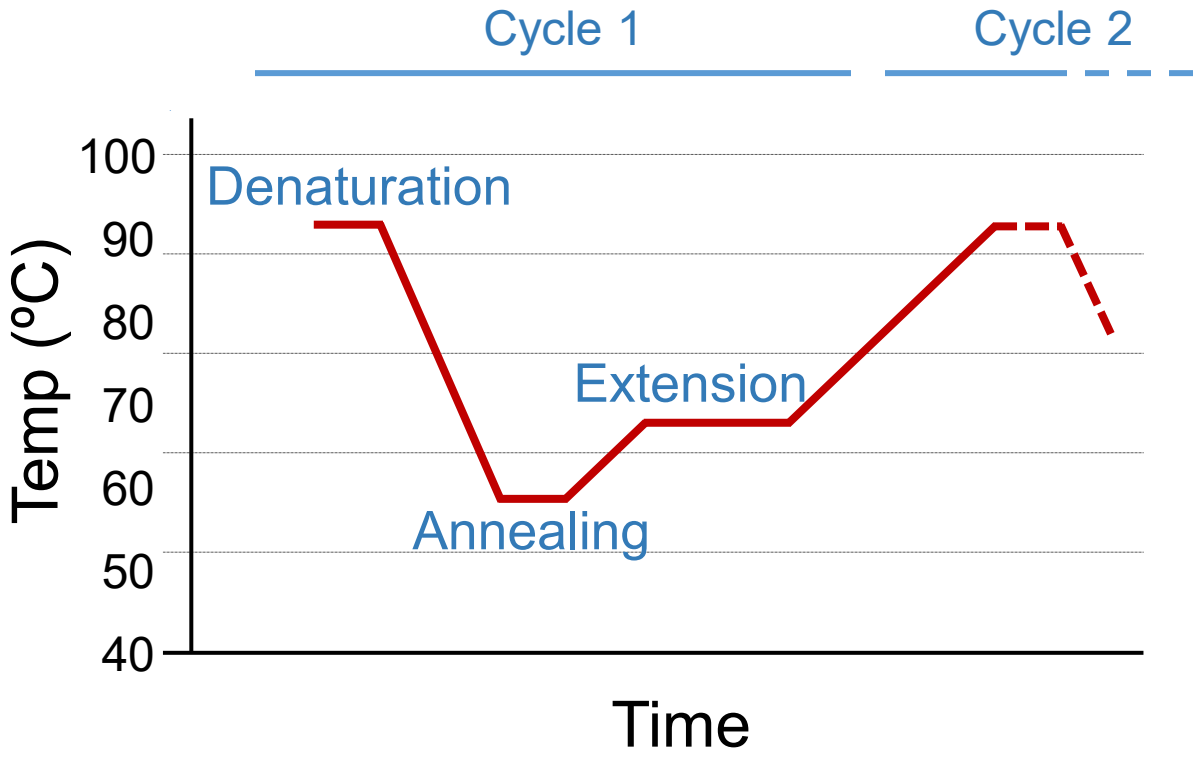
- Sample is cooled down to 50-60 ºC
- Primers bind to their complementary sequences
- This step is critical for specificity

Step 3. Extension

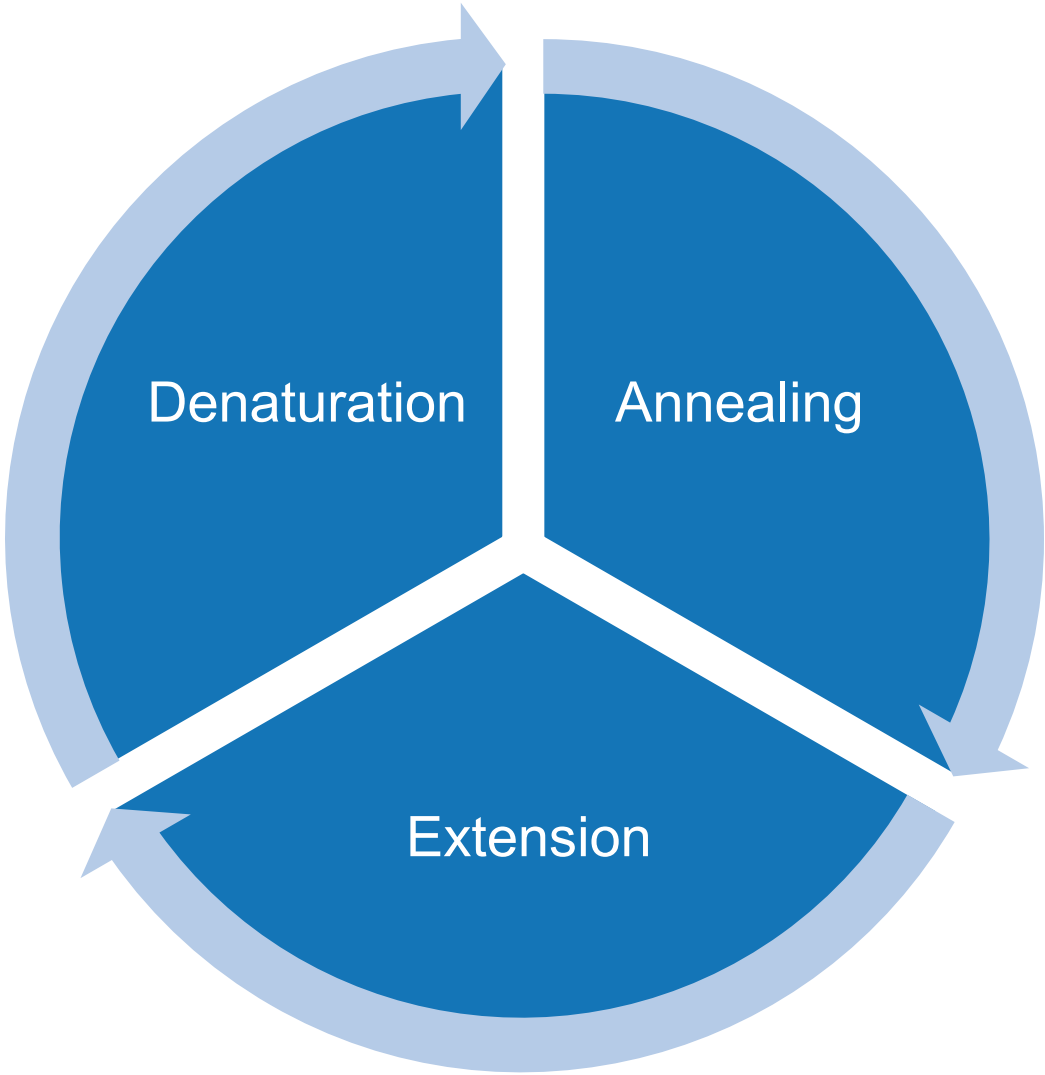


- Sample is heated to 72 ºC
- The single-stranded DNA molecule serves as a template
- DNA polymerase extends each primer from its 3′ end, synthesizing DNA in the 5′ → 3′ direction

Repeated thermal cycles



Cycles



Typically 25-40 cycles, depending on the assay

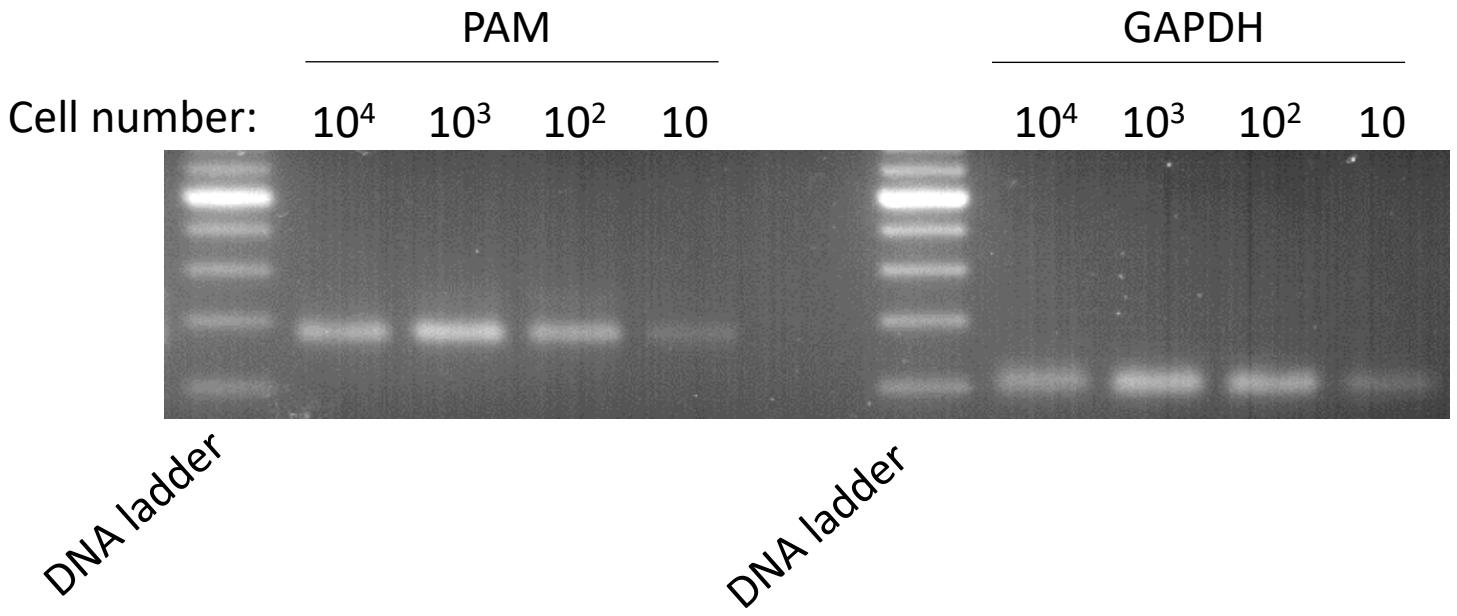
After 35 cycles…

34,359,738,368

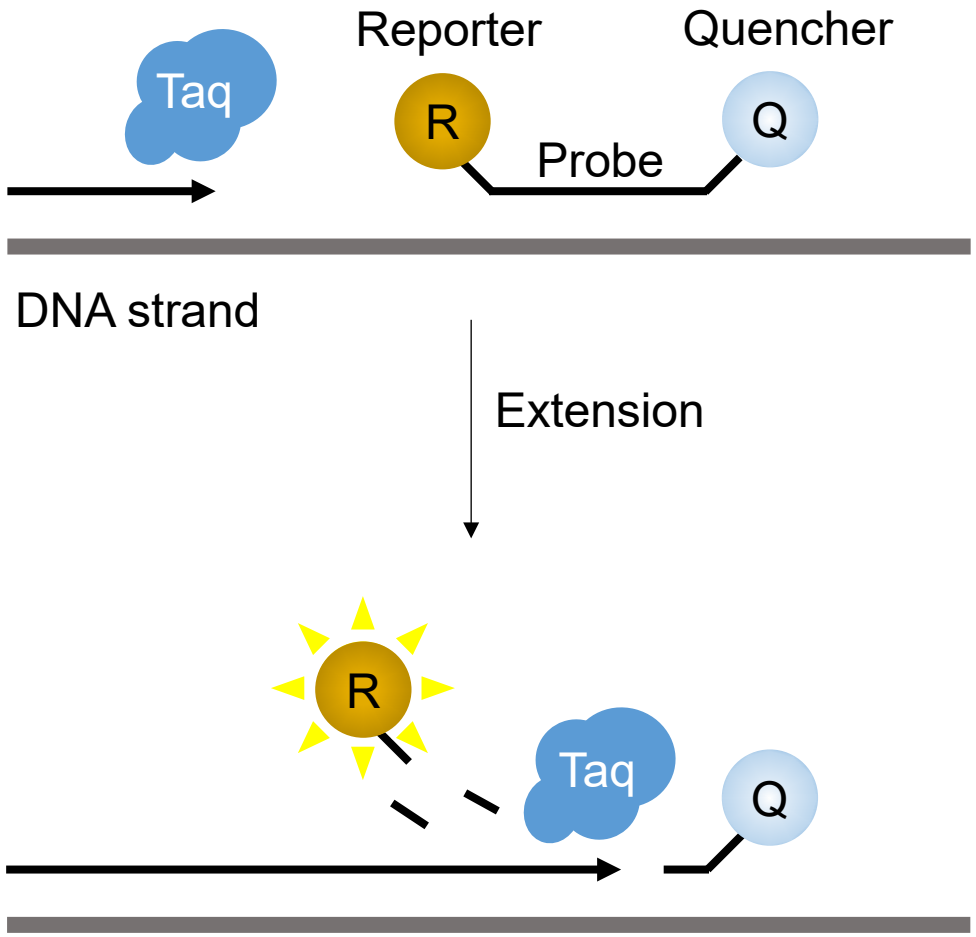
Starting from one DNA molecule and assuming 100% amplification efficiency

Visualization

PCR products are separated by agarose gel electrophoresis, stained with a fluorescent DNA-binding dye such as ethidium bromide, and visualized under UV light



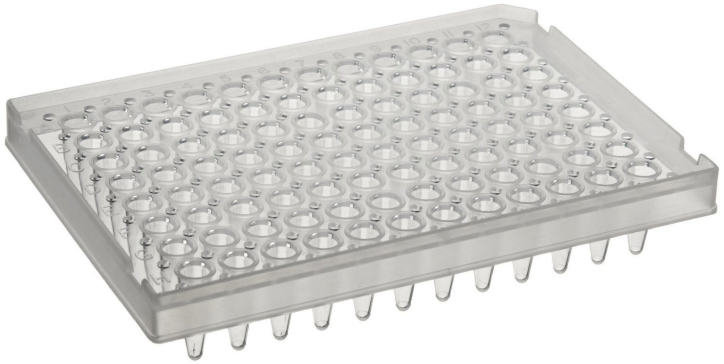
Quantitative PCR (qPCR or real-time PCR)



During extension, Taq polymerase cleaves the probe, separating the reporter from the quencher and increasing fluorescence

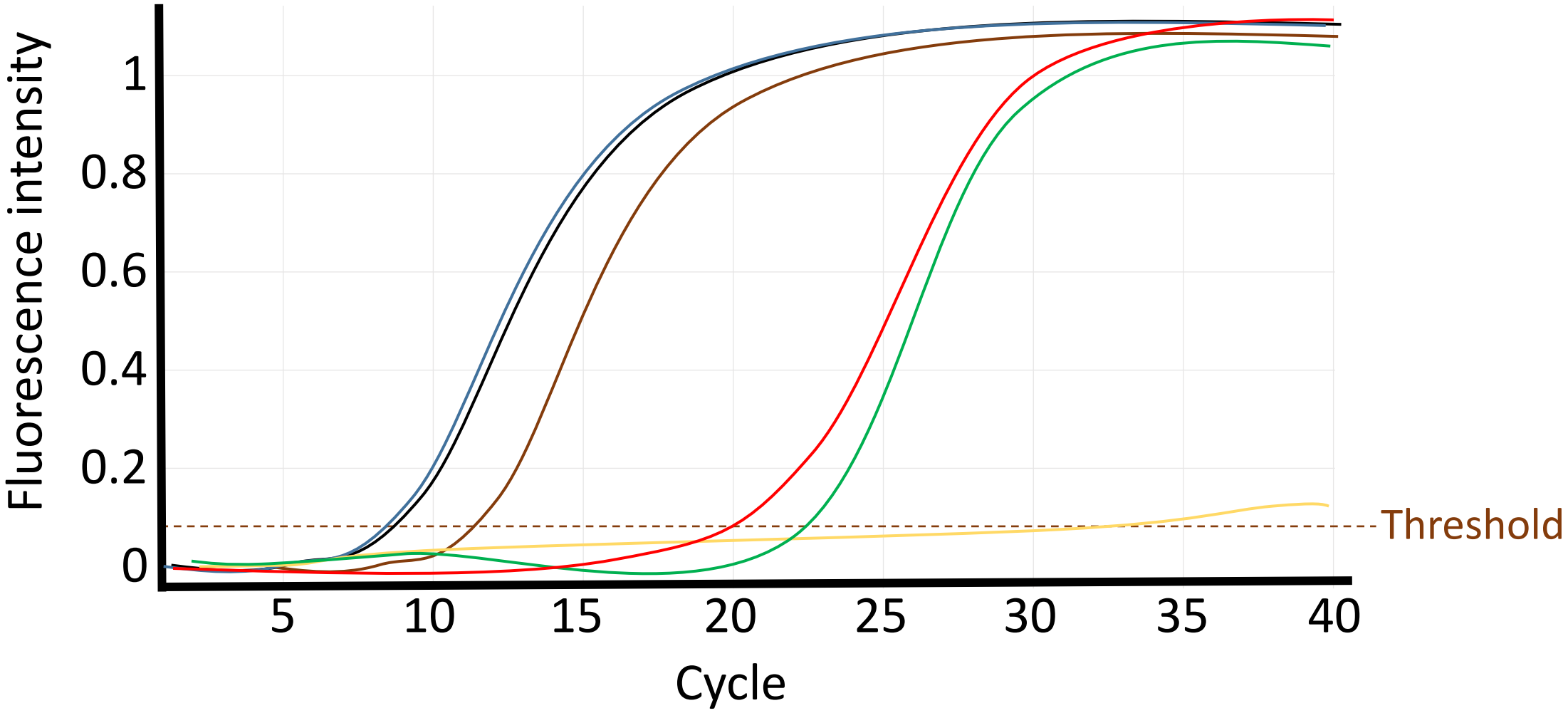
qPCR thermocycler







qPCR visualization

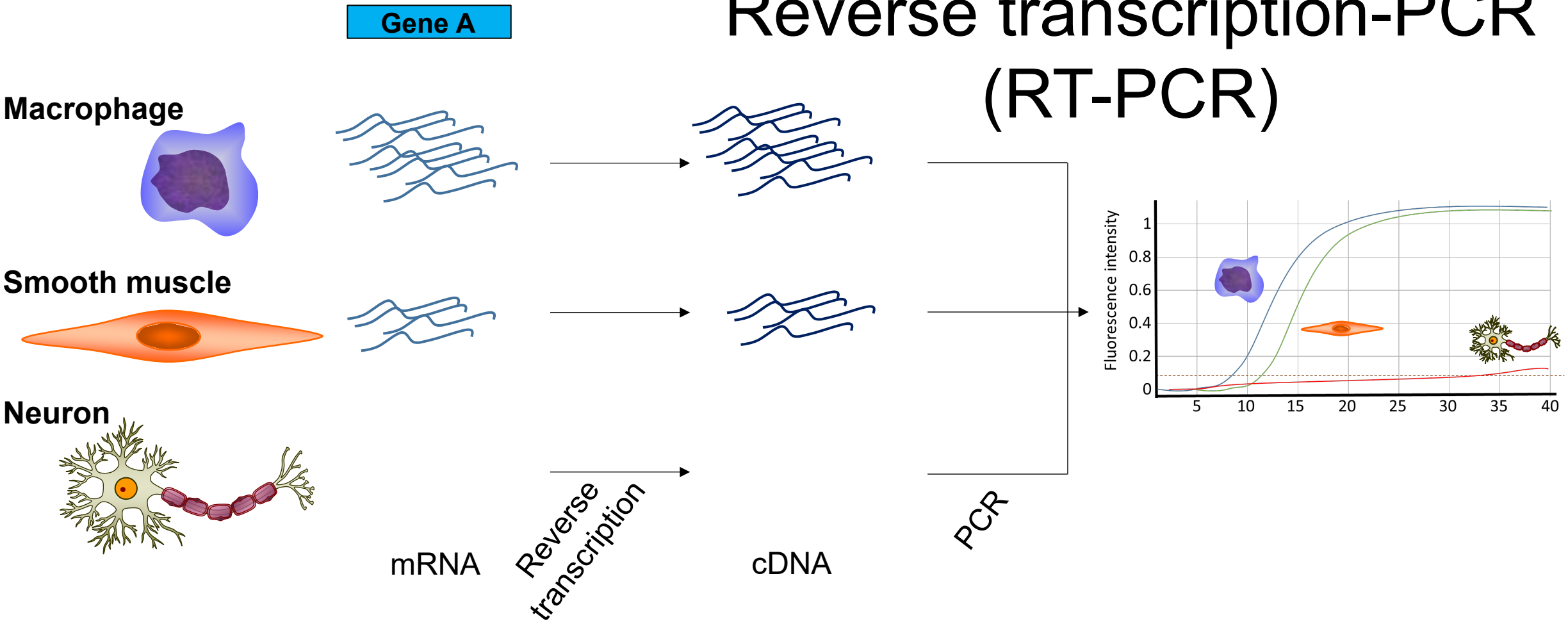


PCR applications

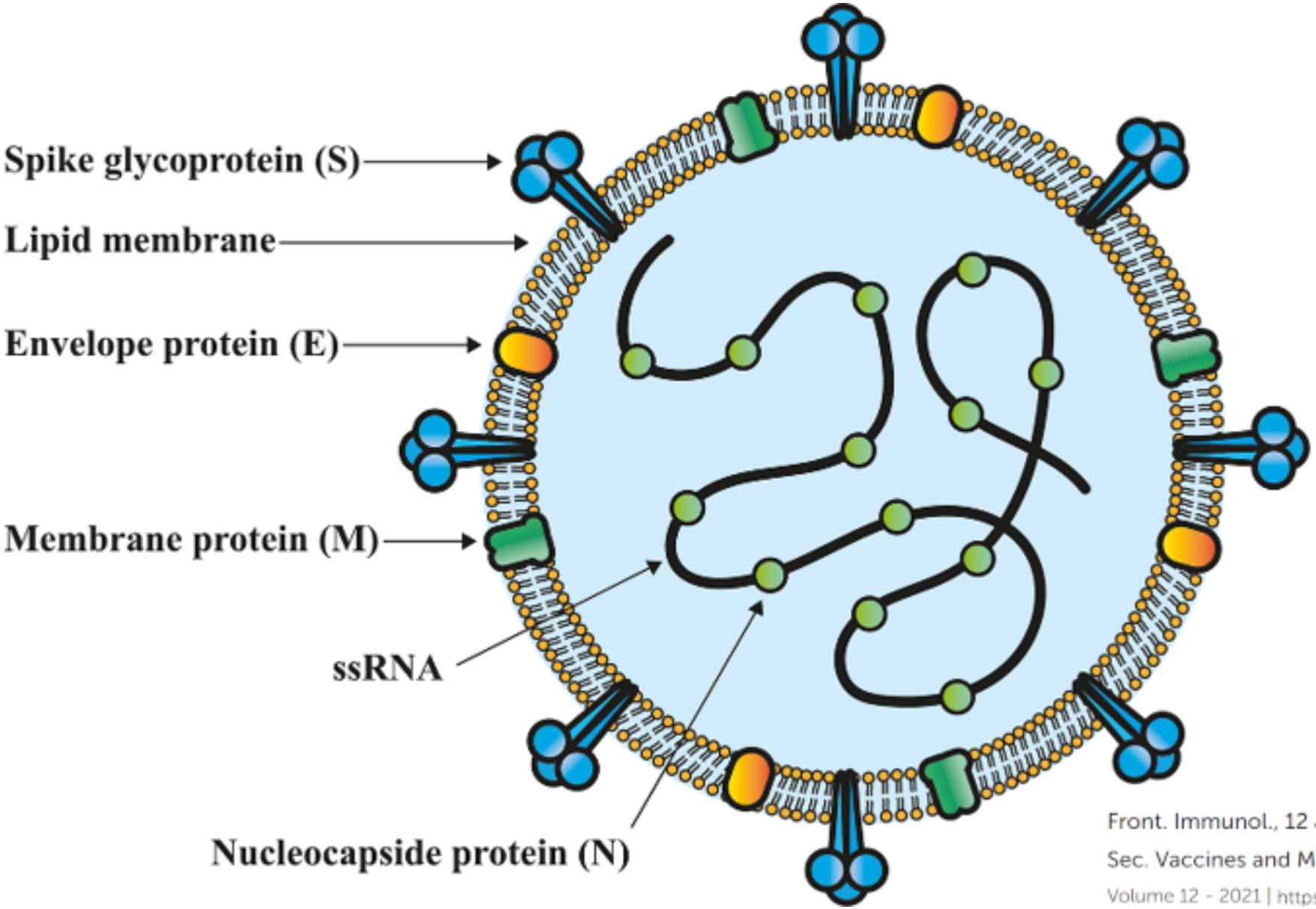
- Molecular identification
    - Gene expression analysis
    - Pathogen detection
    - Forensic medicine
    - Disease-risk identification
    - Disease-treatment customization
- Genetic engineering

Gene expression analysis

Reverse transcription-PCR (RT-PCR)

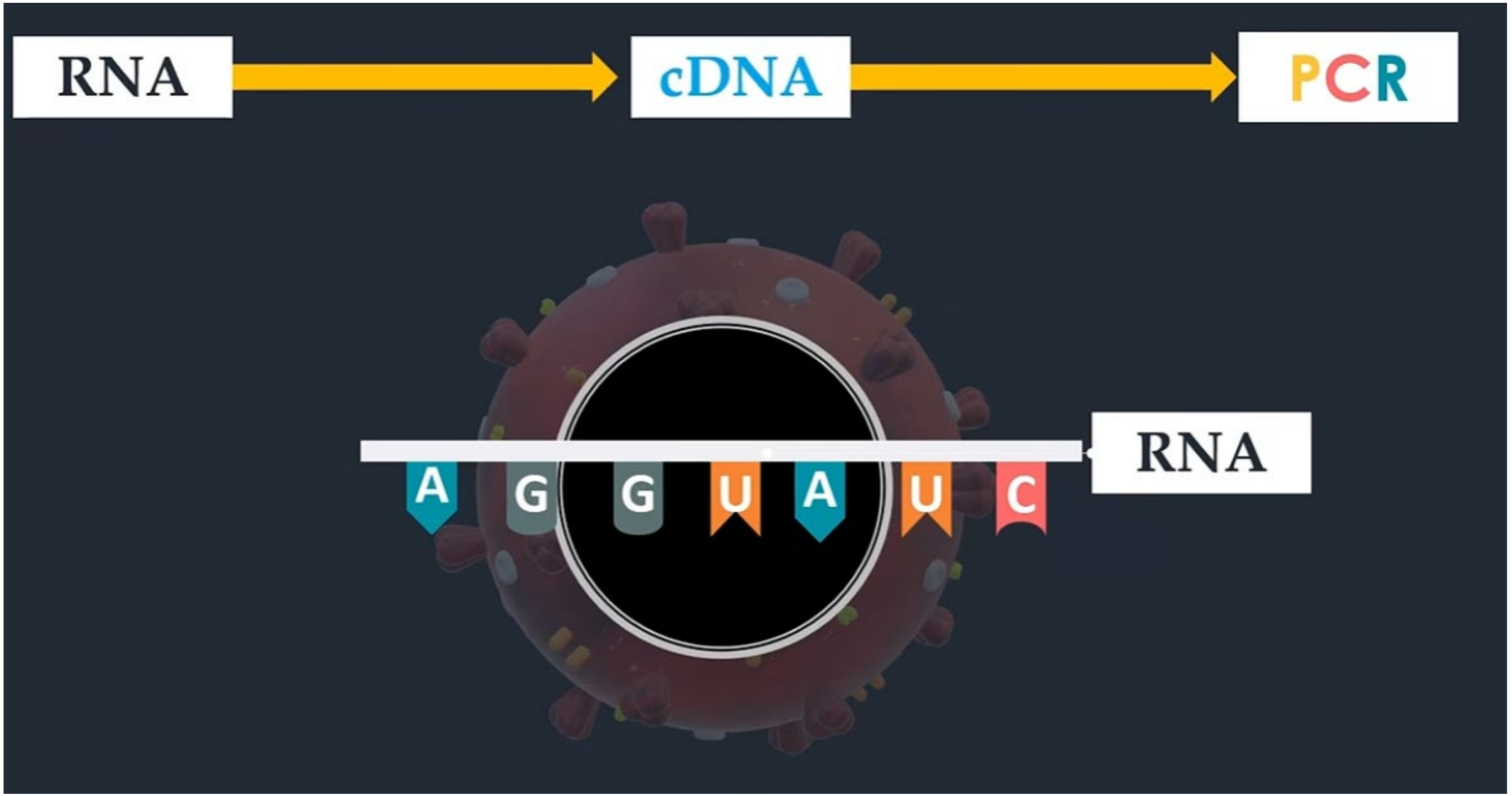


Detection of SARS-CoV-2



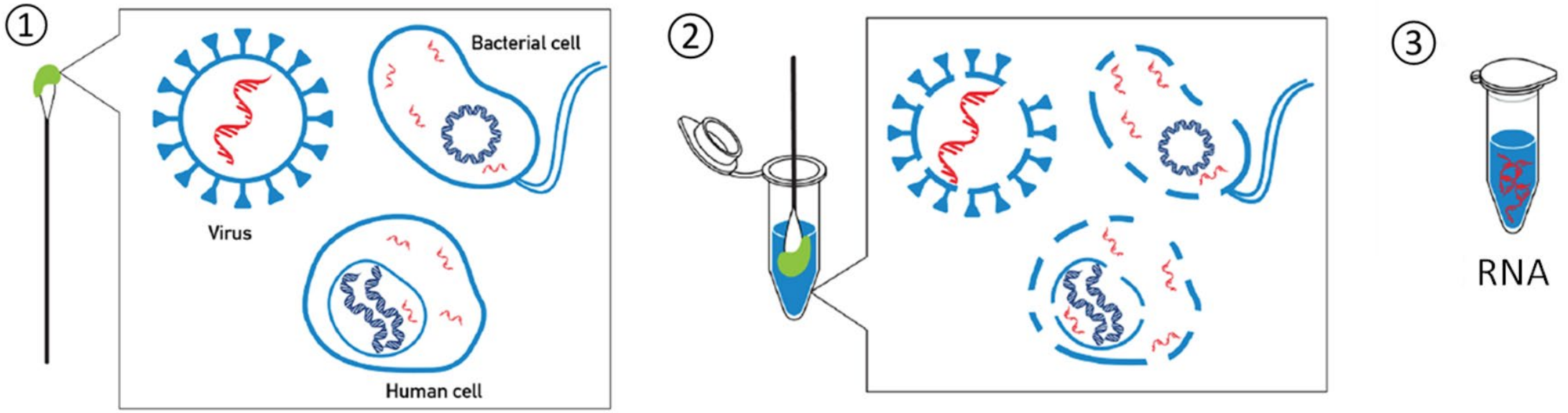
Front. Immunol., 12 July 2021
Sec. Vaccines and Molecular Therapeutics
Volume 12 - 2021 | https://doi.org/10.3389/fimmu.2021.701501

Detection of SARS-CoV-2 by real-time RT-PCR



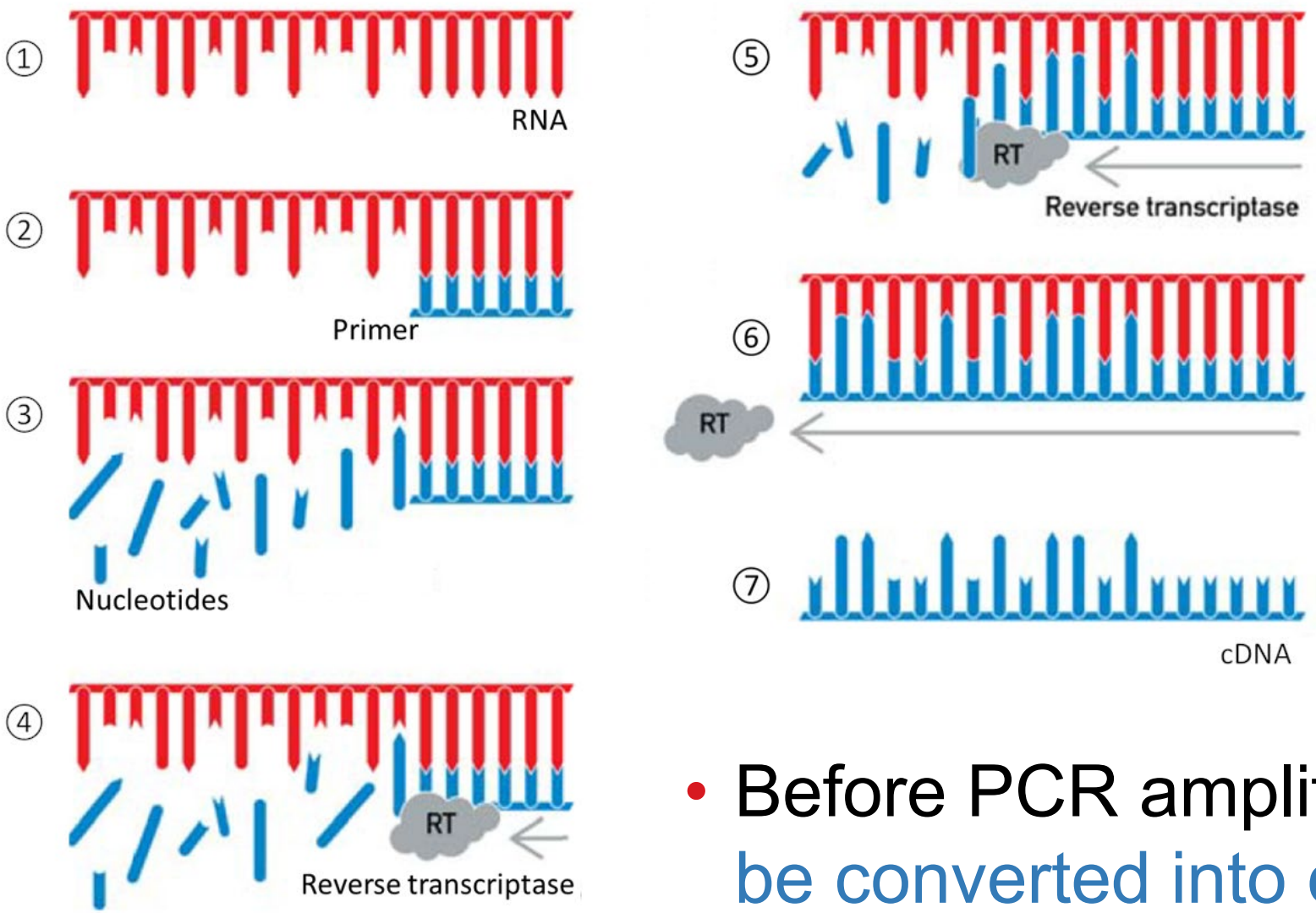
https://www.youtube.com/watch?v=ThG_02miq-4

Step 1. Sample collection and nucleic acid extraction



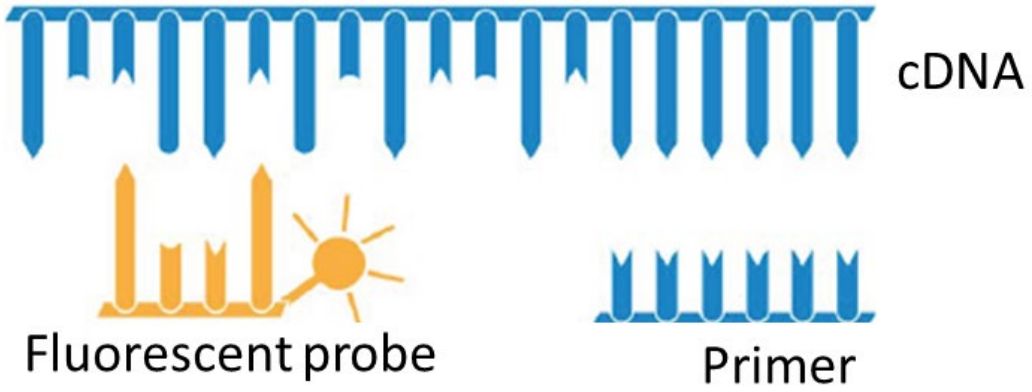
- Obtain patient sample
- Release the genetic material found in the sample
- Purified RNA

Step 2a. Reverse transcription



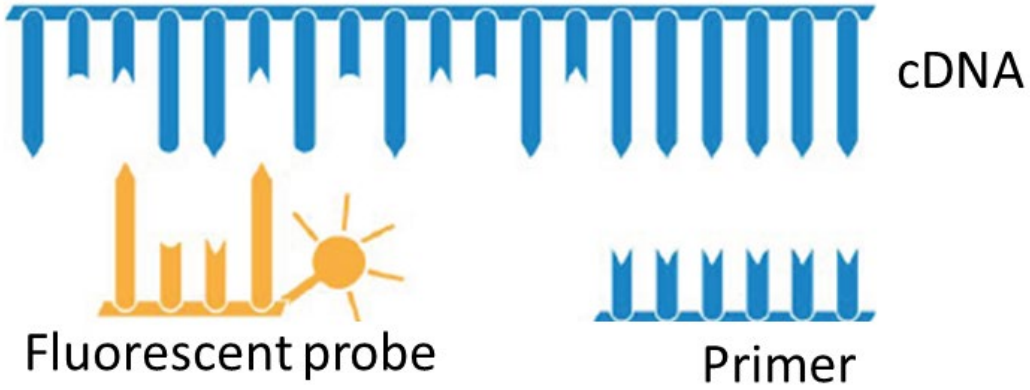
- Before PCR amplification, viral RNA must be converted into cDNA because the Taq polymerase requires a DNA template

Step 2b. PCR



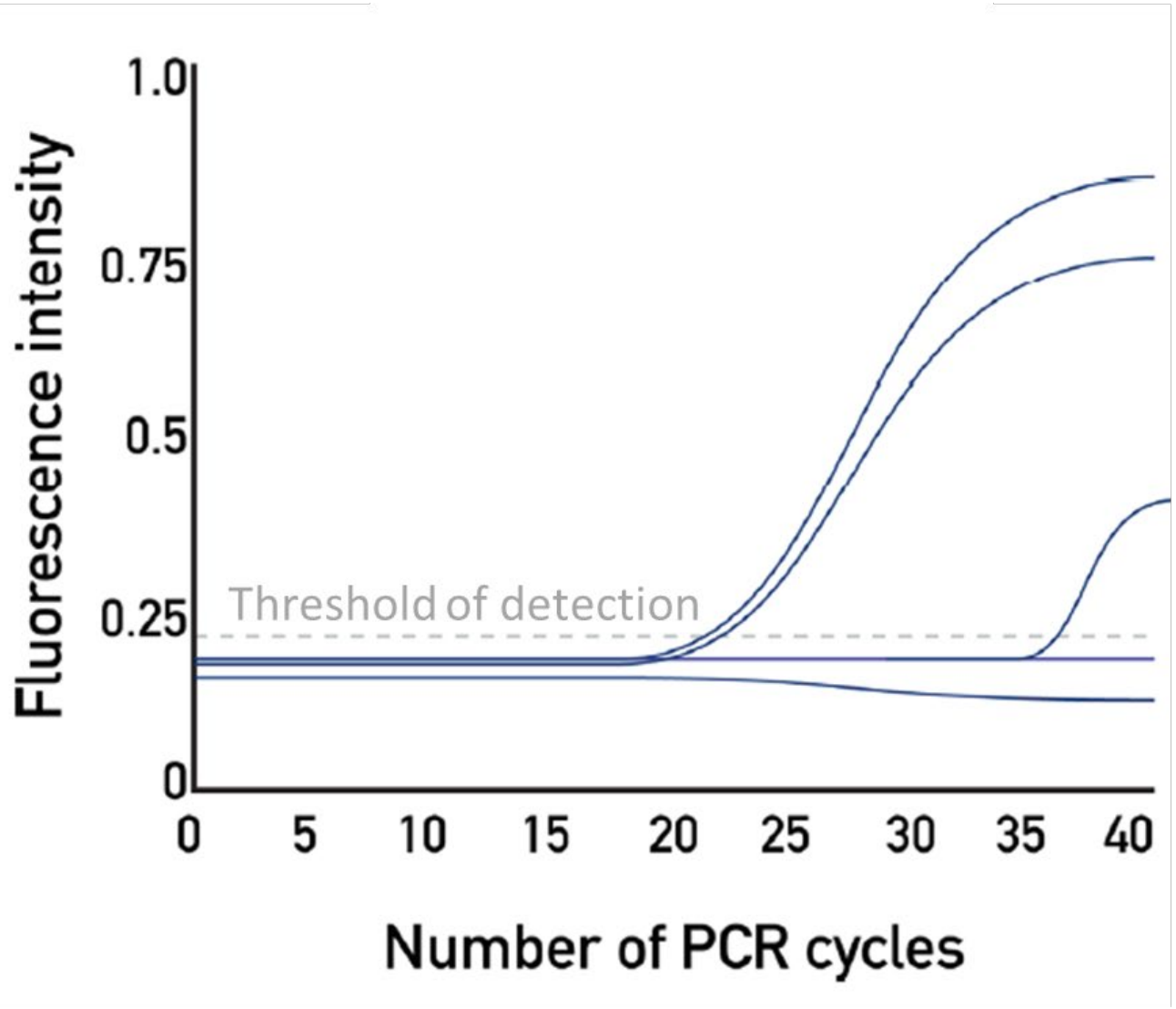
- Search the NCBI database for the SARS-CoV-2 genome: https://www.ncbi.nlm.nih.gov/nuccore/NC_045512.2
- Use NCBI Primer-BLAST to design primers: https://www.ncbi.nlm.nih.gov/tools/primer-blast/

Step 2b. PCR

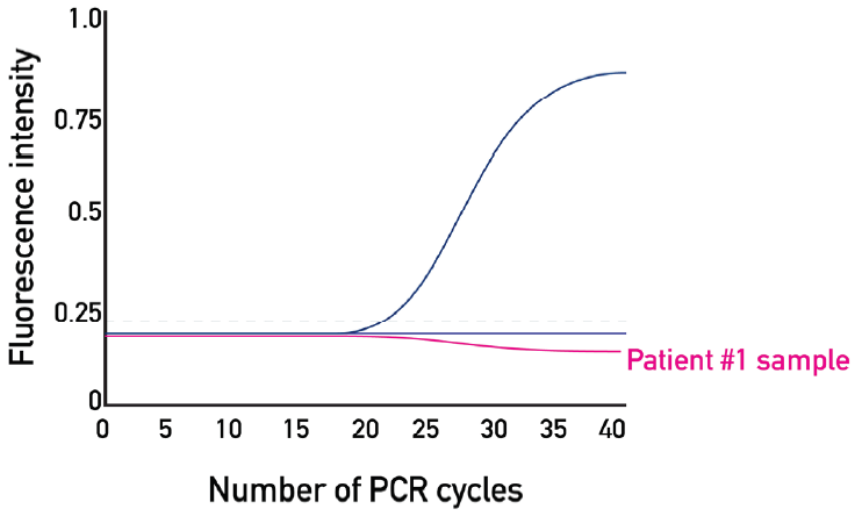


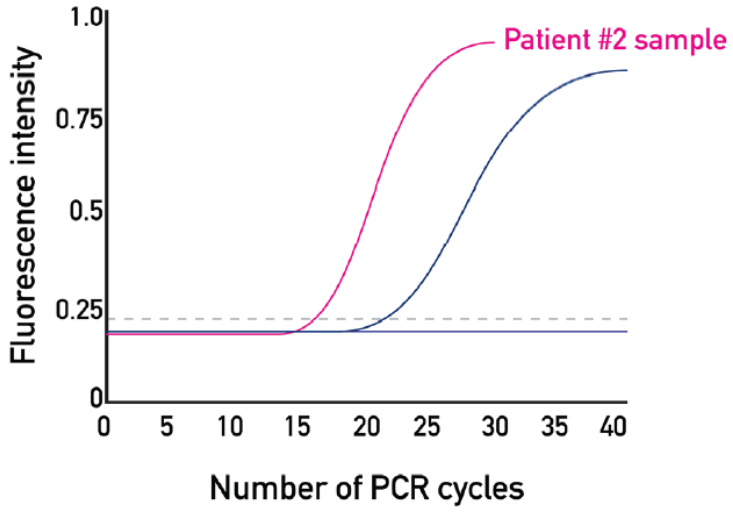
Target	Primer/probe	Sequence (5'-3')
CDC-N1	nCoV_N1-F	GACCCCAAAATCAGCGAAAT
	nCoV_N1-R	TCTGGTTACTGCCAGTTGAATCTG
	nCoV_N1-P	FAM-ACCCCGCATTACGTTTGGTGGACC-IABkFQ
Human RNase P	RP-F	AGATTTGGACCTGCGAGCG
	RP-R	GAGCGGCTGTCTCCACAAGT
	RP-P	Cy5-TTCTGACCTGAAGGCTCTGCGCG-IAbRQ

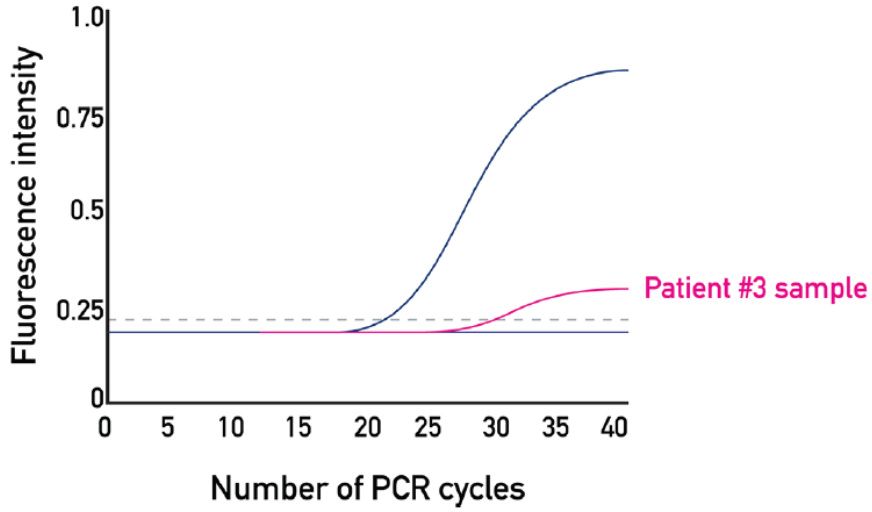
Step 3. Interpretation



Step 3. Interpretation







Patient #1 Result: _____

Patient #2 Result: _____

Patient #3 Result: _____